Leveraging NAIRR for Research: Infrastructure and Hands-On Workshop

Session 1 of 12 — NAIRR Workshop Series

Oakland University & Worcester Polytechnic Institute

March 21, 2026 • 11:00 AM – 2:00 PM ET

Meet the Team

Dr. Darrin Hanna
Professor, Oakland University
Director, High-Performance Embedded Systems Laboratory

- Research focuses on **artificial intelligence** and **embedded systems**
- Contributed to both academic research and practical applications of AI-driven technologies
- Specializes in optimization of computing systems, bridging the gap between hardware and software

Meet the Team

Matthew Horvath

Graduate Research Assistant

High-Performance Embedded Systems Lab, Oakland University

- Pursuing a **Ph.D. in Electrical and Computer Engineering**
- Pattern recognition with data from **automotive sensor systems**
- Safety-critical design automation systems
- Hardware design and testing in **FPGAs**

Meet the Team

Joseph (Joey) Volcic
Graduate Research Assistant
High-Performance Embedded Systems Laboratory, Oakland University

- Lightweight algorithms for **high-speed image processing** using cameras, LIDAR, and thermal sensors
- Experience with **SLAM** (Simultaneous Localization and Mapping) algorithms for autonomous systems
- Enables autonomous systems to map unknown environments while tracking their own location

Today’s Agenda

- Why NAIRR matters for your research (~15 min)
- NAIRR infrastructure deep dive (~15 min)
- Allocation landscape (~10 min)
- Live walkthrough: NAIRR Portal + Jetstream2 (~10 min)

— Hands-On —

- Connecting to Jetstream2 (~15 min)
- Jupyter Notebook Crashcourse (~20 min)
- Exercise A: NAIRR Models + HuggingFace (~25 min)
- Exercise B: Bulk Data Processing at Scale (~25 min)
- Exercise C: NAIRR Open Datasets (~25 min)
- Discussion + Wrap-Up (~10 min)

About This Workshop Series

12 sessions • March–August 2026

Workshop Series Overview

- **12 modular sessions** for researchers at U.S.-based institutions
- Three focus areas: **Cybersecurity**, **Edge AI**, **Autonomous Systems**
- Each session is standalone — attend any combination

What you gain:

- Hands-on experience with national-scale AI infrastructure
- Cross-institutional collaboration (Oakland University + WPI)
- Direct access to NAIRR compute for your ongoing research
- Mentorship and community connections

Workshop Schedule

- **Mar 21** — Intro to NAIRR (today)
- **Apr 4** — Cybersecurity: Threats & Defenses
- **Apr 18** — Edge AI: Deployment & Optimization
- **May 2** — Autonomous Systems: Foundations
- **May 16** — Federated Learning for Privacy
- **May 30** — Adversarial ML & Robustness
- **Jun 13** — Reinforcement Learning in Autonomy
- **Jun 27** — AI for Network Security
- **Jul 11** — Embedded AI for IoT
- **Jul 25** — AI Safety & Alignment
- **Aug 8** — Multi-Agent Systems
- **Aug 22** — Capstone & Showcase

Why NAIRR Matters for Your Research

The compute gap — and how to close it

The Research Compute Bottleneck

- Training models, processing large datasets, and running experiments require compute that **most university labs cannot provide**
- Buying GPU clusters is cost-prohibitive; cloud credits expire; IT departments have constraints
- The result: researchers wait hours for local runs, compromise on dataset size, or abandon compute-intensive approaches entirely

What takes an hour on your laptop might take 2 minutes on Jetstream2.
We will demonstrate this today.

NAIRR: What You Can Access Right Now, for Free

NAIRR is a shared national infrastructure connecting U.S. researchers to:

Computation

- 9 HPC/cloud systems
- NVIDIA A100, H100, L40S GPUs
- Specialized accelerators (Cerebras, Groq, SambaNova)

Data

- AI-ready datasets across scientific domains
- No download fees, no access barriers

Models

- Pre-trained models (NASA, NVIDIA, domain-specific)
- 1,000+ models via Azure AI catalog
- 650+ NVIDIA models on HuggingFace

Support

- This workshop series
- Monthly NAIRR office hours
- AI Unlocked training sessions

Who Runs NAIRR?

- **Led by the National Science Foundation (NSF)**
- Partnership of **14 federal agencies**: DOE, NIH, DARPA, NIST, NASA, NOAA, USDA, and more
- **28 private-sector and nonprofit partners**: Microsoft, NVIDIA, Google, Amazon, OpenAI, HuggingFace, and more

Timeline:

- NAIRR Pilot launched January 2024
- Transitioning to the **NAIRR Operations Center**
- $35 million over 5 years

NAIRR's four objectives: accelerate AI-powered discovery, expand the AI workforce, integrate public + private AI resources, and advance trustworthy AI.

NAIRR's Four Strategic Objectives

1. Accelerate Discovery

- Provide compute and data at the scale modern AI research demands
- Remove infrastructure barriers so researchers focus on the science

2. Expand the AI Workforce

- Train researchers, educators, and students across all institution types
- Ensure access is not limited to elite programs

3. Increase Capacity & Integration

- Connect public and private AI resources into one ecosystem
- Standardize access so switching between systems is straightforward

4. Advance Trustworthy AI

- Promote interpretability, safety, security, and fairness
- Support research into AI risks alongside AI capabilities

NAIRR by the Numbers

600+

Research projects supported

50

U.S. states with active participants

6,000+

Researchers and students served

9

Major compute systems available

This is not a theoretical program. Your peers across the country are actively using NAIRR for research and teaching.

What Researchers Are Doing with NAIRR

Science & Engineering

- Climate modeling with physics-informed neural networks on Delta GPUs
- Drug discovery using protein structure prediction on Bridges-2
- Materials science simulations with high-entropy alloy datasets

Education

- University courses using Jetstream2 classroom allocations for ML assignments
- Workshop series like this one, training researchers across institutions

NLP & Computer Vision

- Fine-tuning large language models on domain-specific corpora
- Satellite image analysis using NVIDIA models from HuggingFace
- Training custom embedding models for scientific search

Interdisciplinary

- Social science researchers analyzing large survey datasets
- Digital humanities projects processing text archives at scale

These are not hypothetical — these projects are actively running on NAIRR systems right now.

Infrastructure Available to You

Compute, data, models, and more

9 Major Compute Systems

- **Jetstream2** (Indiana U) — Cloud VMs, GPUs, classrooms
- **Delta** (NCSA) — 800+ NVIDIA A100 GPUs
- **Bridges-2** (PSC) — GPU + CPU, large memory
- **Anvil** (Purdue) — 1,000+ CPU nodes
- **Frontera** (TACC) — #1 academic supercomputer
- **Lonestar6** (TACC) — A100 GPUs, general purpose
- **Neocortex** (PSC) — Cerebras wafer-scale engines
- **ANL AI Testbed** (Argonne) — Cerebras, Groq, SambaNova
- **ACES** (Texas A&M) — AMD MI250X GPUs

*Today's exercises use **Jetstream2**. The lecture covers what else is available for your larger-scale needs.*

Jetstream2 — Cloud Computing for Research

On-demand virtual machines you control, with GPU options:

Flavor	CPUs	RAM	GPU
m3.small	4	15 GB	—
m3.medium	8	30 GB	—
m3.large	16	60 GB	—
m3.xl	32	125 GB	—
g3.small	8	30 GB	1x NVIDIA A100/H100
g3.large	32	125 GB	1x NVIDIA A100/H100

*Managed via the **Exosphere** portal. Instances take ~3.5 minutes to build. Access via Web Shell, Web Desktop, or SSH tunnel.*

Why Jetstream2 Is the Entry Point

For researchers:

- Full root access — install anything you need
- Persistent instances that stay running between sessions
- GPU flavors (A100, H100) for deep learning
- SSH tunnel workflow — code on your machine, compute on NAIRR
- Attach storage volumes for large datasets

For teaching:

- Classroom allocations give all students identical environments
- No “it works on my machine” issues
- Web Desktop option for participants unfamiliar with CLI
- Scale to any class size without IT bottlenecks
- Students learn on real research infrastructure, not toys

Other NAIRR systems (Delta, Frontera, etc.) require job schedulers and batch queues. Jetstream2 gives you an on-demand VM — the simplest on-ramp to national compute.

Free Models, Datasets, and Azure Credits

NAIRR Free Resources

- nairrpilot.org/pilotresources
- Pre-trained models (some on HuggingFace, some require signup)
- Open datasets across scientific domains
- No allocation needed for many resources

NVIDIA on HuggingFace

- 180+ datasets, 650+ models
- Permissive licenses

Azure AI Model Catalog

- 1,000+ models (OpenAI, Meta, Mistral, etc.)
- $20M credit pool via CloudBank
- First-come, first-served

Pre-installed Software

- PyTorch, TensorFlow, JAX, scikit-learn
- HuggingFace (Transformers, Datasets)
- NVIDIA CUDA/cuDNN/RAPIDS
- Conda/Miniforge, JupyterLab

The HuggingFace Ecosystem — Your Research Toolbox

HuggingFace is tightly integrated with NAIRR. You will use it in all three exercises today.

Datasets Hub

- 200,000+ datasets across every domain
- One line of code to load any dataset
- Text, vision, audio, tabular, scientific
- No registration required for most

Model Hub

- Hundreds of thousands of pre-trained models
- NVIDIA models with permissive licenses
- NASA domain-specific models (we use one today)
- Fine-tune on your own data with `transformers` library

Example — loading a dataset is this simple:

```
dataset = load_dataset("ag_news", split="train")  —  120K articles, ready to use
```

Getting and Managing Allocations

What you can request — and how

Allocation Types

Type	Duration	Best For
Start-Up	3 months	Exploring a new project, prototyping, benchmarking
Research	12 months	Active, funded research with sustained compute needs
Classroom / Education	Semester-length	Teaching AI courses, student assignments, workshops
Deep Partnerships	Varies	Enhanced collaboration with resource providers
Sandbox	Short-term	Quick experimentation, demos

Portal: submit-nairr.xras.org — Submit by the 15th, decision by end of the following month.

You already have ACCESS accounts. If anyone needs help with their account, see the instructor after the session.

What Goes in an Allocation Request

A **Classroom / Education** request is a 3-page PDF (NSF-style formatting):

Course Details

- Institution, course title, enrollment
- Timeline (which semester)
- How AI/compute integrates into the curriculum

Resource Estimates

- GPU vs. CPU needs
- Storage requirements
- Software / frameworks
- Concurrent users expected

Team Qualifications

- PI and co-PI experience
- Prior use of HPC or cloud compute
- Technical readiness

Practical Tips

- Start-Up allocations require less justification — good for getting your feet wet
- Research allocations need a more detailed proposal but give 12 months of access
- Submit by the 15th → decision by end of next month

Info: nairrpilot.org/opportunities/education-call

Today's Allocation

- We are using a **Classroom allocation** on Jetstream2
- Allocation: **NAIRR250048** (Oakland University / WPI)
- Provides: CPU and GPU instances, persistent storage

What this means for today:

- You will connect to a Jetstream2 instance and run exercises there
- Same notebooks run on your laptop *and* on Jetstream2
- Timing comparisons will show the compute advantage

If you want this kind of access for your own research or courses, a Start-Up or Classroom allocation request is straightforward to submit.

Live Walkthrough

The NAIRR Portal and Jetstream2

Portal Tour — Follow Along

Open these in your browser as we walk through them:

1. **nairrpilot.org** — Homepage, Getting Started pathways
2. **submit-nairr.xras.org/resources** — Resource catalog: browse compute, data, models
3. **nairrpilot.org/pilotresources** — Free models & datasets (note the NASA models, HuggingFace links)
4. **Exosphere Dashboard** — Our allocation, instance management

Jetstream2 — How You Connect

Method	What It Is	When to Use
SSH Tunnel	Work from your own machine; Jupyter runs on Jetstream2, appears in your local browser	Day-to-day research workflow (recommended)
Web Desktop	Full Linux desktop (XFCE) in your browser	When you need a GUI or self-contained environment
Web Shell	Terminal in your browser	Quick CLI tasks, package installation
Console	Serial console	Boot/network troubleshooting

Today's exercises use SSH tunnel. Most researchers prefer this — you use your own browser, your own editor, your own shortcuts. The Web Desktop button is available on your instance page if you prefer a GUI.

SSH Tunnel — The Day-to-Day Workflow

Three commands, and Jupyter running on Jetstream2 appears in your local browser:

Step	What You Do	Where
1	`ssh -L 8888:localhost:8888 exouser@<ip>`	Your laptop terminal
2	`conda activate workshop && jupyter lab --no-browser --port=8888`	On the instance (via SSH)
3	Open `http://localhost:8888`	Your laptop browser

What this gives you:

- JupyterLab appears in *your* browser, but code runs on Jetstream2
- Use your own keyboard shortcuts, screen layout, and browser extensions
- Upload/download files through JupyterLab's file browser
- Close your laptop → reconnect later — the instance keeps running

What We'll Do in the Exercises

Exercise	What You'll Build	Key Metric
A: Models + HuggingFace	Semantic search engine using a NASA sentence transformer	Encoding time: 20 → 5,000 passages
B: Bulk Data Processing	Full ML pipeline on 500K rows — features, PCA, grid search	Grid search time across all CPU cores
C: Open Datasets	Text classifier on 120K news articles from HuggingFace	TF-IDF + training time at scale

Every exercise includes a timing comparison table. Run each notebook locally, then run the same notebook on Jetstream2. The difference is the research productivity gain NAIRR provides.

All three notebooks are standalone — if time is short, do any subset. They work on your laptop, on Jetstream2, on Colab, or on any NAIRR system.

Key Takeaways Before We Begin

- **NAIRR gives you free compute, data, and models** for research — no hardware purchases, no expiring cloud credits
- **Jetstream2 is the entry point** — cloud VMs with GPU options, accessible via SSH from your own machine
- **Timing matters:** the exercises will show concrete speedups between your laptop and Jetstream2
- **Allocations are straightforward** — a Start-Up request gets you access in weeks
- **Everything is portable:** the same `.ipynb` notebooks run locally, on Jetstream2, on Colab, or on any NAIRR system

Acknowledgements

- NSF NAIRR-Nat AI Research Resource (NAIRR) Awards OAC-2528533 and OAC-2528534
- National Artificial Intelligence Research Resource (NAIRR) Pilot allocation NAIRR250048
- Jetstream2 cloud resource supported by the NSF (award NSF-OAC 2005506) at Indiana University

Any opinions, finding, and conclusions or recommendations expressed in this material are those of the author(s) and do no necessarily reflect the views of the National Science Foundation.

Contact: fu@oakland.edu | jdai@wpi.edu

Let’s Get Started

First: connect to Jetstream2, then Jupyter crashcourse + exercises